

ARSHDEEP SINGH

Houston, Texas | +1 (281) 818-9955 | Arshdeep.singh@ttu.edu

EDUCATION

Texas Tech University, Lubbock, TX

Expected May, 2027

BS in Computer Engineering | Minor: Mathematics, Robotics

PROJECTS

Autonomous Food Delivery Robot, Project Manager, Lubbock, TX

Dec 2025 - Present

- Leading bring-up and integration of **perception, localization, and control subsystems** on an **NVIDIA Jetson** platform, focusing on sensor synchronization, noise reduction, and real-time performance for future autonomous navigation
- Coordinating **ROS2-based development** integrating **LiDAR** and vision sensors, debugging actuator control and timing issues in preparation for precise obstacle avoidance at **target speeds up to 1.5 m/s**
- Designed **onboarding processes and technical documentation**, streamlining knowledge transfer for **15+ team** members across semesters

3D LiDAR Scanning System, Independent Project, Lubbock, TX

November 2025 – December 2025

- Built custom **3D scanner** converting **2D LiDAR** via rotating platform, generating accurate point clouds at **96% lower cost** (under **\$200 total vs. \$5000+** commercial 3D Sensors)
- Developed **1,300 lines of Python and C++** code for motion control and synchronization, reducing reconstruction errors via precise calibration
- Designed **SolidWorks CAD** and mechanical assembly, overcoming alignment challenges to produce high-resolution scans
- Built **Point Cloud** visualization pipeline to validate spatial accuracy of captured data

Humanoid Robot ("SpaceDroid"), Team Lead

May 2024

Inter-School & College Science Competition (2nd Place)

- Directed **3-member team** to design and build a **half-humanoid robotic system** by integrating **Raspberry Pi, Camera, IMU Sensors, and Servo** joints in a unified control system
- Implemented **OpenCV object-detection** and **voice interaction** in Python and C++, enabling responsive **software-driven actuation** for complex tasks
- Resolved perception-to-motion translation challenges, achieving reliable **path planning** and joint coordination in competitive demonstration

LEADERSHIP EXPERIENCE

Robotics and Advanced Tech Society, Project Manager, Lubbock, TX

Aug 2024 - Present

- Onboarded **20+ members** across semesters, accelerating project ramps by ~50% through structured guidance on **electrical and software problems**
- Coordinated **multi-level teams** on **autonomous prototypes**, troubleshooting bottlenecks to deliver **function demos** ahead of deadlines

Raider Aerospace Society, Systems Team Lead, Lubbock, TX

Aug 2023 – April 2025

- Led electronics and software development for AIAA aircraft, creating custom **LoRa based monitoring tools**, achieving **sub-second telemetry latency** and stable **communication across 200-500m test ranges** during test flights
- Oversaw **autonomous drone from 200ft release**, attaining **~95% landing accuracy** within **5m** target zones via refined control algorithms

Texas Tech Polo Club, Operations Manager, Lubbock, TX

May 2025 – Dec 2025

- **Increased club's budget by ~56%** by negotiating a sponsorship agreement with **Mercedes-Benz of Lubbock**
- Managed day-to-day operations, logistics, safety, and maintenance of **10+ acre of facility and 9+ horses**
- Represented the men's team in **varsity-level competitions**

SKILLS

- *Languages & Tools:* Python, C, C++, C#, Java, Linux, Bash, Git
- *Robotics & Embedded:* ROS2, NVIDIA Jetson, Raspberry Pi, LiDAR, Arduino, SLAM, Sensor Fusion
- *Simulation & Design:* Unity3D, SolidWorks (CAD), Altium PCB Designer, Gazebo, CARLA
- *AI & Perception:* Deep Learning (ANN, CNN), Object Detection, Dataset Handling, Model Validation, Debugging ML Outputs